Dear Hiring Manager,

I am writing to express my interest in opportunities at L3Harris Technologies, particularly in software engineering, computer vision, machine learning, or systems development.

I am a Mechanical Engineer and Software Developer with over six years of experience at Hatch, where I worked at the intersection of engineering, industrial software, computer vision, edge computing, and real-time systems.

One of the areas I have found most compelling throughout my career is applying computer vision and machine learning to real-world environments where reliability, performance, and integration with physical systems are critical.

At Hatch, I developed and deployed computer vision solutions for mining applications, including camera-based inspection and analysis using YOLO-based models. I worked with Python, PyTorch, Flask, Angular, and GPU-accelerated technologies to develop systems capable of processing visual information for industrial applications.

This work extended beyond model development. I was involved in deploying machine-learning systems onto edge computing infrastructure, where processing data close to the source was important for operational applications. This gave me practical experience with the challenges of taking computer vision from an experimental environment into a deployable system, including data ingestion, networking, compute constraints, model deployment, APIs, databases, and system integration.

I have also developed VISP, a personal computer vision platform designed around deploying YOLO-based applications on edge devices. The platform supports RTSP cameras, detection storage and querying, and higher-level applications such as alerts, analytics, and operational dashboards. Building this system required me to work across computer vision, networking, databases, backend services, and infrastructure rather than treating the machine-learning model as an isolated component.

In addition to computer vision, my professional experience includes developing and maintaining real-time industrial software. At Hatch, I contributed to VISTA, a hydro energy plant operating and optimization system developed using C++ and C#. I worked on data ingestion and performance-analysis components while maintaining and modernizing legacy C++ software. This experience strengthened my understanding of real-time systems, performance considerations, software architecture, and the integration of software with industrial control and automation systems.

My engineering background is another important part of what I would bring to L3Harris. As a Mechanical Engineer, I have worked on large-scale industrial projects and spent time in commissioning and field environments. This has taught me to design and develop technology with the realities of physical systems, operational constraints, reliability, and end users in mind.

I am particularly interested in L3Harris because of the opportunity to work on sophisticated systems where software, sensing, computer vision, embedded computing, and physical hardware come together. I would be excited to apply my experience developing computer vision and edge-computing systems while continuing to deepen my expertise in high-performance and real-time software development.

Thank you for considering my application. I would welcome the opportunity to discuss how my experience in computer vision, machine learning, edge computing, real-time systems, and engineering could contribute to L3Harris Technologies.

Sincerely,

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